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Limits and Colimits

Cones and Cocones

Cone over $F: J \rightarrow C$: Object X with compatible morphisms to $F(j)$ **Cocone under $F: J \rightarrow C$:** Object X with compatible morphisms from $F(j)$

Limit = Terminal Cone

Lim F : Universal cone with unique morphisms from any other cone

Colimit = Initial Cocone

Colim F : Universal cocone with unique morphisms to any other cocone

15 Examples

1. **Product:** Limit of discrete diagram
2. **Coproduct:** Colimit of discrete
3. **Equalizer:** Limit of parallel pair
4. **Coequalizer:** Colimit of parallel
5. **Pullback:** Limit of span diagram
6. **Pushout:** Colimit of cospan
7. **Kernel:** Equalizer with zero
8. **Cokernel:** Coequalizer with zero
9. **Terminal object:** Limit of empty diagram
10. **Initial object:** Colimit of empty
11. **Inverse limit:** Limit of inverse system
12. **Direct limit:** Colimit of direct system
13. **Fiber product:** Pullback
14. **Amalgamated free product:** Pushout
15. **Projective/Inductive limits**

Preservation

Right adjoints preserve limits Left adjoints preserve colimits

Key property for understanding adjunctions!

Complete/Cocomplete Categories

- **Complete:** Has all limits
- **Cocomplete:** Has all colimits
- **Bicomplete:** Both

Examples: Set, Top, Grp, Vect, Presheaves (all bicomplete)

Completeness Theorem

Theorem: $\mathcal{C}$ is complete iff has products and equalizers **Dual:** $\mathcal{C}$ is cocomplete iff has coproducts and coequalizers

(Other limits/colimits build from these)

Universal Properties

Limits/colimits characterized by universal properties Foundation for categorical algebra and Kan extensions

Study Guide

- Master products and coproducts
- Understand pullbacks deeply
- Work through examples
- Connect to adjoints
- Recognize in your field